

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Physics — SSC-I (9th Grade)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. Instrument for measuring time to 0.1 s precision: ✓ **A — Digital stopwatch** 1 mark

The textbook states: "Generally the **least count of analogue stopwatch is 1 s and digital stop watch is 0.1 s.**" Since the digital stopwatch has the smaller least count, it can resolve time intervals to 0.1 s, ten times finer than the analogue (mechanical) stopwatch.

Q2. Least count of vernier caliper (0.5 mm main scale, 25 vernier divisions): ✓ **B — 0.02 mm** 1 mark

Least Count = Smallest division on main scale ÷ Total divisions on vernier scale = $0.5 \text{ mm} \div 25 = 0.02 \text{ mm}$. This is finer than the standard 1 mm/10-division caliper ($LC = 0.1 \text{ mm}$), since both the smaller main-scale division and the larger number of vernier divisions increase precision.

Q3. Acceleration of motorboat (8 m/s → 26 m/s in 6 s): ✓ **C — 3 m/s²** 1 mark

Using **$a = (v - u)/t$** = $(26 - 8)/6 = 18/6 = 3 \text{ m/s}^2$. Option B (18 m/s^2) is the change in velocity without dividing by time; Option D (4.33 m/s^2) mistakenly divides the final velocity by time instead of the change in velocity.

Q4. Time to reach maximum height ($u = 25 \text{ m/s}$, $g = 10 \text{ m/s}^2$): ✓ **D — 2.5 s** 1 mark

At maximum height, final velocity $v = 0$. Using **$v = u - gt$** : $0 = 25 - 10t$, so **$t = 25/10 = 2.5 \text{ s}$** . This is the time for the upward-moving ball's velocity to be brought to zero by gravitational deceleration.

Q5. Acceleration of trolley ($F = 32 \text{ N}$, $m = 8 \text{ kg}$): ✓ **A — 4 m/s²** 1 mark

By **Newton's second law, $F = ma$** , so **$a = F/m$** = $32 \text{ N} \div 8 \text{ kg} = 4 \text{ m/s}^2$. Option C (40 m/s^2) comes from adding $32 + 8$ instead of dividing; Option D (256 m/s^2) from multiplying; Option B (0.25) inverts the ratio.

Q6. Forces in the FBD of a book resting on a table: ✓ **C — Weight and the Normal force from the table** 1 mark

A **free-body diagram (FBD)** shows only the forces acting *on* the object being analysed. A book at rest on a table (with nothing pushing it sideways) has exactly two forces: its **weight (F_g)** acting downward, and the **Normal force (F_N)** from the table pushing upward. Since the book is in equilibrium, these two forces are equal and opposite. No friction acts because there is no applied force trying to slide the book.

Q7. Centre of gravity of an L-shaped (bent) uniform lamina: ✓ **D — May lie outside the lamina, in empty space** 1 mark

The textbook explains (Figure 4.7) that the CM/CG "doesn't always lie inside the mass... depending upon the mass distribution... For example, if the arm is bent, the CM/CG shifts its location to the new location and **may lie outside the mass distribution.**" An L-shaped lamina is a classic case where the geometric centre falls in the empty notch of the "L", outside the actual material.

Q8. Torque ($F = 25 \text{ N}$, $d = 0.3 \text{ m}$): ✓ **A — 7.5 N·m** 1 mark

Torque = Force × perpendicular distance = $\tau = F \times d = 25 \text{ N} \times 0.3 \text{ m} = 7.5 \text{ N·m}$. Option D (75 N·m) results from a decimal-point error.

Q9. Pressure of 0.76 m mercury column: ✓ **B — $1.03 \times 10^5 \text{ Pa}$** 1 mark

Using $P = \rho gh = 13,600 \text{ kg/m}^3 \times 10 \text{ m/s}^2 \times 0.76 \text{ m} = 103,360 \text{ Pa} \approx 1.03 \times 10^5 \text{ Pa}$. This is essentially how the standard atmospheric pressure value is obtained from a mercury barometer reading of 0.76 m (760 mm) of mercury.

Q10. Power of pump ($W = 3000 \text{ J}$, $t = 20 \text{ s}$): ✓ **C — 150 W** 1 mark

Power = Work done ÷ time, $P = W/t = 3000 \text{ J} \div 20 \text{ s} = 150 \text{ W}$. Power is the **time rate of doing work**, measured in watts ($1 \text{ W} = 1 \text{ J/s}$).

Q11. Energy resource from heat trapped in Earth's crust by radioactive decay: ✓ **A — Geothermal** 1 mark

The textbook (Section 6.3.6) states: "**Geothermal energy** comes from the Earth's core and is formed by the trapped energy from billions of years ago and the heat produced by **decaying radioactive elements**."

Biomass comes from plants, tidal from the Moon's gravitational pull, and wave from surface ocean waves — none involve the Earth's internal heat.

Q12. Branch of physics studying blood flow and respiration: ✓ **D — Biophysics** 1 mark

The textbook (Section 9.2) states: "**Biophysics** is the study of dynamics of **blood flow and respiration**, radiation in diagnostics and treatment etc." Astrophysics studies celestial bodies, Geophysics studies the Earth's structure, and Acoustics studies sound — none deal with human body systems.

SECTION B — Short Questions Answer Key ($11 \times 3 = 33$ Marks)

2(i) — Physical balance: construction and use 3 marks

A **physical balance** consists of a vertical pillar supporting a horizontal beam that rests on a knife edge, with a pan hanging from each end. A long **pointer** is attached to the middle of the beam, which swings over a scale so its resting point can be read. **Levelling screws** at the base are used to level the instrument, and it is enclosed in a protective glass case to keep dust and air currents from disturbing the reading.

1.5 marks

Use: The body to be weighed is placed in one pan, and known standard weights (from the weight box supplied with the balance) are placed in the other pan until the pointer settles back at the centre of the scale, showing the beam is balanced. The total of the standard weights used then gives the mass of the body. A physical balance is sensitive enough to measure mass to the **milligram** order. 1.5 marks

OR

Method: A **measuring cylinder** (graduated cylinder) is partly filled with water and the initial water level is noted. The irregular solid is then lowered in so that it is fully submerged, and the new (higher) water level is noted. Since the solid displaces its own volume of water, the **volume of the solid = final reading — initial reading**. 1.5 marks

Initial water level = 24 mL Final water level = 63 mL Volume of stone = $63 \text{ mL} - 24 \text{ mL} = 39 \text{ mL} = 39 \text{ cm}^3$

Volume of the stone = 39 cm^3 1.5 marks

2(ii) — Scientific notation and SI prefixes 3 marks

(a) $0.000085 \text{ m} = 8.5 \times 10^{-5} \text{ m}$. Using the prefix **micro** (μ) = 10^{-6} : $8.5 \times 10^{-5} \text{ m} = 85 \times 10^{-6} \text{ m} = 85 \text{ }\mu\text{m}$ (**micrometres**). 1.5 marks

(b) $52,000,000 \text{ s} = 5.2 \times 10^7 \text{ s}$. 1.5 marks

OR**SI prefixes:** 1.5 marks

Multiplier	Prefix	Symbol
10^{-2}	centi	c
10^6	mega	M
10^9	giga	G

Conversion: $250 \text{ mm} = 250 \times 10^{-3} \text{ m} = 0.25 \text{ m} = 2.5 \times 10^{-1} \text{ m}$ 1.5 marks

2(iii) — Uniform acceleration numerical 3 marks

(a) $u = 6 \text{ m/s}$, $v = 26 \text{ m/s}$, $t = 8 \text{ s}$. Using $a = (v - u)/t$:

$$a = (26 - 6) / 8 = 20 / 8 = 2.5 \text{ m/s}^2$$

Acceleration = 2.5 m/s^2 1.5 marks

(b) Using $s = ut + \frac{1}{2}at^2$:

$$s = (6)(8) + (0.5)(2.5)(8^2) \quad s = 48 + (0.5)(2.5)(64) \quad s = 48 + 80 = 128 \text{ m}$$

Distance covered = 128 m 1.5 marks

OR

Uniform acceleration: velocity changes by **equal amounts in equal time intervals** (e.g. a car speeding up steadily on a straight highway). **Non-uniform acceleration:** velocity changes by **unequal amounts** in equal time intervals (e.g. a car in stop-start city traffic). 1.5 marks

Deceleration: $u = 40 \text{ m/s}$, $v = 0$, $t = 8 \text{ s}$.

$$a = (v - u)/t = (0 - 40)/8 = -5 \text{ m/s}^2$$

The negative sign shows the velocity is decreasing; **deceleration = 5 m/s^2** . 1.5 marks

2(iv) — Distance and displacement 3 marks

(a) Total path length travelled (distance is scalar, direction ignored):

$$\text{Distance} = 500 \text{ m} + 200 \text{ m} = 700 \text{ m}$$

Total distance = 700 m 1 mark

(b) Displacement is measured from the starting point to the final position, along the straight road. Since the jogger runs 500 m forward and then 200 m back (in the opposite direction), her net position is:

$$\text{Displacement} = 500 \text{ m} - 200 \text{ m} = 300 \text{ m (towards her original direction of travel)}$$

Magnitude of displacement = 300 m 2 marks

OR

The **slope (gradient) of a speed-time graph gives the acceleration** of the body. The **area under a speed-time graph gives the distance travelled** by the body. 1 mark

For constant speed, the graph is a horizontal line, so the "area" is a simple rectangle:

$$\text{Distance} = \text{speed} \times \text{time} = 15 \text{ m/s} \times 8 \text{ s} = 120 \text{ m}$$

Distance covered = 120 m 2 marks

2(v) — Newton's second law numerical 3 marks

Newton's second law: The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass, acting in the direction of the net force: $\mathbf{F} = m\mathbf{a}$. 1 mark

(a) $F = 42 \text{ N}$, $m = 7 \text{ kg}$:

$$a = F/m = 42 \text{ N} / 7 \text{ kg} = 6 \text{ m/s}^2$$

Acceleration = 6 m/s² 1 mark

(b) Same force, $m = 3.5 \text{ kg}$:

$$a = F/m = 42 \text{ N} / 3.5 \text{ kg} = 12 \text{ m/s}^2$$

New acceleration = 12 m/s² (halving the mass doubles the acceleration for the same force). 1 mark

OR

Inertia: the natural tendency of a body to resist any change in its state of rest or uniform motion. The greater the mass of a body, the greater its inertia. 1.5 marks

Carpet-beating example: When a hanging carpet is struck with a stick, the carpet itself is forced to move suddenly, but the dust particles trapped within it are not directly struck — by **inertia**, they tend to remain at rest. As the carpet moves away from under them, the dust particles are left behind (unable to follow the sudden motion) and fall away, which is why beating a carpet removes dust. 1.5 marks

2(vi) — Impulse and force from a hockey strike 3 marks

(a) $m = 0.16 \text{ kg}$, $u = 0$, $v = 30 \text{ m/s}$. Change in momentum:

$$\Delta p = m(v - u) = 0.16 \text{ kg} \times (30 - 0) \text{ m/s} = 4.8 \text{ kg}\cdot\text{m/s}$$

Change in momentum = 4.8 kg·m/s 1.5 marks

(b) Using $F = \Delta p / \Delta t$, with $\Delta t = 0.02 \text{ s}$:

$$F = \Delta p / \Delta t = 4.8 \text{ kg}\cdot\text{m/s} / 0.02 \text{ s} = 240 \text{ N}$$

Average force exerted by the stick = 240 N 1.5 marks

OR

Law of conservation of momentum: in the absence of an external force, the total momentum of a system remains constant. For the boy + bag system (initially at rest, total momentum = 0): 1 mark

$$\begin{aligned} 0 &= (\text{mass of boy})(\text{recoil velocity}) + (\text{mass of bag})(\text{velocity of bag}) \\ 0 &= (45 \text{ kg})(v) + (3 \text{ kg})(6 \text{ m/s}) \\ v &= -(3 \times 6)/45 = -0.4 \text{ m/s} \end{aligned}$$

(a) **Recoil speed of the boy = 0.4 m/s** (in the direction opposite to the bag). 1 mark

(b) Kinetic energy of the boy:

$$KE = \frac{1}{2}mv^2 = 0.5 \times 45 \text{ kg} \times (0.4 \text{ m/s})^2 = 0.5 \times 45 \times 0.16 = 3.6 \text{ J}$$

Kinetic energy of boy = 3.6 J 1 mark

2(vii) — See-saw balance (principle of moments) 3 marks

For balance, the clockwise moment must equal the anticlockwise moment about the pivot: 1 mark

$$\text{Rabia's moment} = \text{Adeel's moment} \quad (300 \text{ N})(0.8 \text{ m}) = (400 \text{ N})(d) \quad 240 \text{ N}\cdot\text{m} = 400 \text{ N} \times d \quad d = 240 / 400 = 0.6 \text{ m}$$

Adeel must sit 0.6 m from the pivot. 2 marks

OR

Principle of moments: a body is in rotational equilibrium when the sum of clockwise moments about a pivot equals the sum of anticlockwise moments. 1 mark

$$\begin{aligned} \text{Distance of 9 N weight from pivot (40 cm mark)} &= 40 - 10 = 30 \text{ cm} \\ \text{Moment of 9 N weight} &= 9 \text{ N} \times 30 \text{ cm} = 270 \text{ N}\cdot\text{cm} \\ \text{Let the 6 N weight be hung at distance } d \text{ (cm) from the pivot on the other side: } &6 \text{ N} \times d = 270 \text{ N}\cdot\text{cm} \\ d &= 270 / 6 = 45 \text{ cm} \\ \text{Mark position} &= 40 \text{ cm} + 45 \text{ cm} = 85 \text{ cm mark} \end{aligned}$$

The 6 N weight must be hung at the 85 cm mark. 2 marks

2(viii) — Centre of gravity and stability 3 marks

The **position of the centre of gravity (CG)** of a body determines how easily it topples. A body remains stable as long as a vertical line dropped from its CG falls within its base of support; if the body is tilted enough that this line falls outside the base, an unbalanced moment is created that topples it further (unstable equilibrium). 1.5 marks

A bus has a **high CG** because much of its mass (passengers, body) is located well above the road. When rounding a bend, the bus experiences a sideways tilt; because its CG is high, only a small tilt is needed before the vertical line through the CG moves outside its wheelbase, so it becomes easy to overturn. A car has a **low CG** (heavy engine and chassis close to the ground), so it must tilt through a much larger angle before the CG line moves outside its base — making it far more stable when cornering. 1.5 marks

OR

The **centre of gravity (CG)** of a body is the point at which the entire weight of the body may be considered to act. 1 mark

Experimental method for an irregular lamina:

1. Make a small hole near the edge of the lamina and suspend it freely from a pin so it can swing.
2. Hang a plumb line (a string with a small weight) from the same pin, and mark the line it traces on the lamina.
3. Repeat this from at least **two more different points** on the lamina, each time marking the vertical line traced by the plumb line.
4. The point where all these lines intersect is the **centre of gravity** of the lamina. 2 marks

2(ix) — Frictional dissipation 3 marks

A **dissipative force** is a force (such as friction or air resistance) that always opposes the motion of an object and converts its mechanical energy into other forms, usually **heat**, so that the total mechanical energy of the system decreases. 1 mark

Meteor example: As a meteor enters the Earth's atmosphere at very high speed, it compresses and rubs against the air in front of it. The **friction between the meteor and the air molecules** dissipates its kinetic energy as heat, causing the surrounding air (and the meteor itself) to heat up intensely. This heat is so great that most meteors vaporise completely, producing the visible glowing streak we call a "shooting star." 2 marks

OR

When two hands are rubbed together, the rough surfaces of the skin resist relative motion — this resistance is **friction**. Rubbing does **work against friction**, and this work is **dissipated as heat**, which is why the hands feel warmer. 1.5 marks

Energy conversion: the mechanical energy used in moving the hands back and forth is converted into **thermal (heat) energy**. 1.5 marks

2(x) — Liquid pressure numerical 3 marksUsing $P = \rho gh$: 1 mark

$$P = \rho gh = (800 \text{ kg/m}^3)(10 \text{ m/s}^2)(4 \text{ m}) \quad P = 32,000 \text{ Pa}$$

Pressure at the bottom of the tank = 32,000 Pa ($3.2 \times 10^4 \text{ Pa}$) 2 marks**OR**

A **manometer** is a device used to measure the pressure of a gas or liquid, usually consisting of a U-shaped tube containing a liquid. Its working principle is that the **difference in height between the liquid levels in the two limbs** is directly proportional to the pressure difference being measured, given by $P = \rho gh$.

1.5 marks

$$P = \rho gh = (900 \text{ kg/m}^3)(10 \text{ m/s}^2)(0.25 \text{ m}) \quad P = 2250 \text{ Pa}$$

Pressure difference indicated = 2250 Pa 1.5 marks**2(xi) — Pressure numerical ($P = F/A$)** 3 marks

Pressure is defined as the force acting normally (perpendicularly) per unit area: $P = F/A$. Its SI unit is the **pascal (Pa)**, where $1 \text{ Pa} = 1 \text{ N/m}^2$. 1 mark

$$\text{Force (weight of tractor) } F = mg = 2500 \text{ kg} \times 10 \text{ m/s}^2 = 25,000 \text{ N} \quad P = F/A = 25,000 \text{ N} / 0.5 \text{ m}^2 = 50,000 \text{ Pa}$$

Pressure exerted on the soil = 50,000 Pa ($5 \times 10^4 \text{ Pa}$) 2 marks**OR**

Since $P = F/A$, for a given force, pressure **decreases** as the area of contact **increases**, and **increases** as the area **decreases**. A tractor is fitted with **wide tyres** to spread its weight over a large contact area, reducing the pressure on soft ground so it does not sink in. A nail has a **sharp pointed tip** to concentrate the applied force on a tiny area, producing a very large pressure that allows it to penetrate wood easily.

2 marks

Another example: a rucksack with wide, padded straps reduces the pressure on the shoulders compared to a thin strap carrying the same weight. 1 mark

SECTION C — Long Questions Answer Key ($4 \times 5 = 20$ Marks)

3(i) — Equations of motion: derivation + numerical **5 marks**

(a) Derivation of $s = ut + \frac{1}{2}at^2$: On a speed-time graph for a body accelerating uniformly from u to v in time t , the distance travelled equals the **area under the graph**, which is a trapezium split into a rectangle (base t , height u) and a triangle (base t , height $v - u = at$): **2 marks**

$$s = \text{area of rectangle} + \text{area of triangle} \quad s = (u)(t) + \frac{1}{2}(t)(v - u) \quad \text{Since } v - u = at: \quad s = ut + \frac{1}{2}(t)(at) \\ s = ut + \frac{1}{2}at^2$$

(b) $u = 12 \text{ m/s}$, $a = 2 \text{ m/s}^2$, $t = 6 \text{ s}$.

$$(i) \quad v = u + at = 12 + (2)(6) = 12 + 12 = 24 \text{ m/s} \quad (ii) \quad s = ut + \frac{1}{2}at^2 = (12)(6) + (0.5)(2)(6^2) \quad s = 72 + (0.5)(2)(36) = 72 + 36 = 108 \text{ m}$$

Velocity after 6 s = 24 m/s; Distance covered = 108 m **2 marks**

(c) Check using $v^2 = u^2 + 2as$: **1 mark**

$$v^2 = u^2 + 2as = 12^2 + 2(2)(108) = 144 + 432 = 576 \quad v = \sqrt{576} = 24 \text{ m/s} \quad \checkmark \text{ (matches part (b)(i))}$$

OR

(a) Three equations of motion: **2 marks**

Equation	Symbols
$v = u + at$	u = initial velocity, v = final velocity, a = acceleration, t = time, s = distance covered
$s = ut + \frac{1}{2}at^2$	
$2as = v^2 - u^2$	

(b) Stone dropped from rest: $u = 0$, $g = 10 \text{ m/s}^2$, $t = 5 \text{ s}$.

$$(i) \quad h = ut + \frac{1}{2}gt^2 = 0 + (0.5)(10)(5^2) = (0.5)(10)(25) = 125 \text{ m} \quad (ii) \quad v = u + gt = 0 + (10)(5) = 50 \text{ m/s}$$

Height of tower = 125 m; Velocity at impact = 50 m/s **3 marks**

3(ii) — Conservation of momentum + kinetic energy loss **5 marks**

(a) The **law of conservation of momentum** states that in the absence of an external force, the total momentum of a system before collision equals the total momentum after collision. For two bodies that collide and move together with common velocity v : **1 mark**

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

(b) $m_1 = 1000 \text{ kg}$, $u_1 = 15 \text{ m/s}$; $m_2 = 500 \text{ kg}$, $u_2 = 0$ (stationary):

$$(1000)(15) + (500)(0) = (1000 + 500)v \quad 15,000 = 1500v \quad v = 15,000 / 1500 = 10 \text{ m/s}$$

Common velocity after collision = 10 m/s **2 marks**

(c) Kinetic energy before and after collision:

$$KE(\text{before}) = \frac{1}{2}(1000)(15^2) = 0.5 \times 1000 \times 225 = 112,500 \text{ J} \quad KE(\text{after}) = \frac{1}{2}(1500)(10^2) = 0.5 \times 1500 \times 100 \\ = 75,000 \text{ J} \quad \text{Loss in KE} = 112,500 - 75,000 = 37,500 \text{ J}$$

Kinetic energy lost = 37,500 J (this energy is converted into sound, heat and deformation of the vehicles during the collision) **2 marks**

OR

(a) Impulse is the product of the net force acting on a body and the time for which it acts: **Impulse = $F\Delta t$** . Derivation from Newton's second law: **2 marks**

$$F = ma = m(v - u)/t = (mv - mu)/t = \Delta p/\Delta t \therefore F\Delta t = \Delta p \quad (\text{Impulse} = \text{change in momentum})$$

(b) $m = 1.2 \text{ kg}$, $u = 12 \text{ m/s}$, $v = 0$, $\Delta t = 0.008 \text{ s}$:

$$(i) \quad \text{Impulse} = \Delta p = m(v - u) = 1.2 \times (0 - 12) = -14.4 \text{ kg}\cdot\text{m/s} \quad \text{Magnitude of impulse} = 14.4 \text{ N}\cdot\text{s} \quad (ii) \quad F = \Delta p/\Delta t = 14.4 / 0.008 = 1800 \text{ N}$$

Impulse = 14.4 N·s; Average force on hand = 1800 N **3 marks**

3(iii) — Liquid pressure: derivation + layered-liquid numerical 5 marks

(a) Derivation of $P = \rho gh$: Consider a liquid column of cross-sectional area A and height h , density ρ . Its volume is $V = Ah$, so its mass is $m = \rho Ah$, and its weight is $F = mg = \rho Ahg$. This weight acts on the base of area A , so the pressure at the base is: 2 marks

$$P = F/A = (\rho Ahg)/A = \rho gh$$

(b) Oil layer: $\rho = 800 \text{ kg/m}^3$, $h = 0.5 \text{ m}$; water layer: $\rho = 1000 \text{ kg/m}^3$, $h = 2.5 \text{ m}$; $g = 10 \text{ m/s}^2$.

$$(i) P(\text{oil}) = \rho gh = (800)(10)(0.5) = 4000 \text{ Pa} \quad (ii) P(\text{bottom}) = P(\text{oil}) + \rho(\text{water}) \cdot g \cdot h(\text{water}) \\ P(\text{bottom}) = 4000 + (1000)(10)(2.5) \quad P(\text{bottom}) = 4000 + 25,000 = 29,000 \text{ Pa}$$

Pressure at oil-water interface = 4000 Pa; Total pressure at bottom = 29,000 Pa 3 marks

OR

(a) A simple mercury barometer is made by filling a long glass tube (closed at one end) completely with mercury, then inverting it into an open trough of mercury, without letting air enter. The mercury level in the tube falls slightly, leaving a vacuum at the closed top, until the column is supported by the atmospheric pressure acting on the mercury in the trough. The **vertical height of the mercury column represents the atmospheric pressure** at that place — a taller column means higher atmospheric pressure. 2 marks

(b) $h = 0.70 \text{ m}$, $\rho = 13,600 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$:

$$P = \rho gh = (13,600)(10)(0.70) = 95,200 \text{ Pa}$$

Atmospheric pressure at the hill station = 95,200 Pa. Since $95,200 \text{ Pa} < 1.013 \times 10^5 \text{ Pa}$, this is **lower** than the sea-level value — consistent with the fact that atmospheric pressure decreases with increasing altitude. 3 marks

3(iv) — Conservation of energy: numerical + energy resources 5 marks

(a) The **law of conservation of energy** states that energy can neither be created nor destroyed; it can only be converted from one form to another, and the total energy of an isolated system remains constant.

1 mark

(b) $m = 4 \text{ kg}$, $u = 20 \text{ m/s}$, $g = 10 \text{ m/s}^2$:

$$(i) KE = \frac{1}{2}mv^2 = 0.5 \times 4 \times 20^2 = 0.5 \times 4 \times 400 = 800 \text{ J} \quad (ii) \text{ At maximum height (no friction), all KE converts to PE: } mgh = KE \quad h = KE / (mg) = 800 / (4 \times 10) = 800/40 = 20 \text{ m}$$

Kinetic energy at projection = 800 J; Maximum height (no air resistance) = 20 m 3 marks

(c) Actual maximum height = 18 m:

$$PE(\text{actual}) = mgh = 4 \times 10 \times 18 = 720 \text{ J} \quad \text{Work done against air resistance} = KE - PE(\text{actual}) = 800 - 720 = 80 \text{ J}$$

Work done against air resistance = 80 J 1 mark

OR

(a) 2 marks

Renewable	Non-renewable
Naturally replenished over time; cannot run out (e.g. geothermal heat from the Earth's core, tidal energy from the Moon's pull)	Limited in amount, formed over millions of years, cannot be replaced once used (e.g. coal, natural gas, petroleum)
Other examples: biomass (energy stored in plants via photosynthesis), wind energy	Other example: nuclear fuel (uranium)

(b) $m = 2000 \text{ kg}$ (falling each second), $h = 15 \text{ m}$, $g = 10 \text{ m/s}^2$:

$$(i) PE \text{ lost per second} = mgh = 2000 \times 10 \times 15 = 300,000 \text{ J} \quad (ii) \text{ Power} = \text{Energy} / \text{time} = 300,000 \text{ J} / 1 \text{ s} = 300,000 \text{ W} = 300 \text{ kW}$$

PE lost per second = 300,000 J ($3 \times 10^5 \text{ J}$); Power output = 300 kW 3 marks